

# Education, Lifestyles, and Health Inequality

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## INTRODUCTION

- Health inequalities in the United States are large
- Two important facts:
  - a) Strong connection between economic and health inequality  
Kitagawa, Hauser (1973); Pijoan-Mas, Rios-Rull (2014); Chetty et al. (2016)
  - b) Growing educational gradients of health inequality  
Preston, Elo (1995); Meara et al. (2009); Montez et al. (2011); Case, Deaton (2015)
- Reasons not well-understood
- We study to which extent differences in health behaviors across education groups can account for these two facts
  - Health behaviors are an important driver of health outcomes  
McGinnis, Foege (1993); Li et al. (2018); Zaninotto et al. (2020)
  - Individuals with higher education are more likely to adopt health-enhancing behaviors  
Lantz et al. (1998); Cutler, Lleras-Muney (2010)

## OBJECTIVES

- ① Measure the impact of lifestyles on health dynamics and economic outcomes
  - Lifestyle → propensity to engage in different health-related behaviors
- ② Understand the joint determination of education and lifestyles
  - Why is there an education gradient of lifestyles?
    - a) Do higher educated individuals have higher incentives to adopt protective lifestyles?
    - b) Do individuals raised in healthier environments self-select into higher education?
- ③ Understand the increase in the education gradient of life expectancy
  - Quantify the role played by the increase in the education wage premium

## WHAT WE DO: DATA

- There are many indicators of **health behavior** in HRS and PSID  
(*preventive tests, substance abuse, exercise*)
  - Problems
    - They are imperfectly correlated across individuals and over time
    - Hard to identify their long-run effect on health outcomes
  - We identify patterns in **health behavior types** that we interpret as **lifestyles** based on:
    - Cross-sectional and panel variation of health behaviors
    - Health dynamics
- Types are permanent, but health behaviors change with age and health status

## WHAT WE DO: MODEL

- Two different stages
    - 2) Adulthood — standard life-cycle model with labor income and health risks
      - Education and lifestyle given
      - Choice of consumption and savings
      - Provides *early life* value of each education and lifestyle choice
    - 1) Early life — simple static problem
      - Heterogeneous costs of acquiring education and lifestyle
      - Once and for all choice of education and lifestyle
      - Provides joint distribution of education and lifestyle
  - **Complementarities** between education and lifestyle choices
    - + An extra year of life is more valuable with higher consumption possibilities
- ⇒ **Selection**: individuals from healthier backgrounds tend to pursue higher education.

## Health Dynamics and Health Behavior

## THE DATA

- HRS and PSID
  - Unbalanced panels of individuals  $i = 1, \dots, N$  followed  $t = 1, \dots, T$  periods
- Demographic information: birth cohort ( $c$ ), sex ( $s$ ), education ( $e$ ), age of entry ( $a_0$ )
- Wide array of information on **health status** and **health behavior**
  - Health state  $h_t$ : self-reported health (good/bad) + death
  - Health behavior  $z_{mt} \in \{0, 1\}$ :
    - Preventive cancer tests (mammography / prostate check)
    - Cholesterol test
    - Flu shot
    - Heavy drinking (2+ drinks every day)
    - Smoking
    - Exercise
  - Across demographic groups: associated w/ health outcomes
  - At individual level: imperfectly correlated across individuals and over time

[\[See data\]](#)

## LATENT TYPES

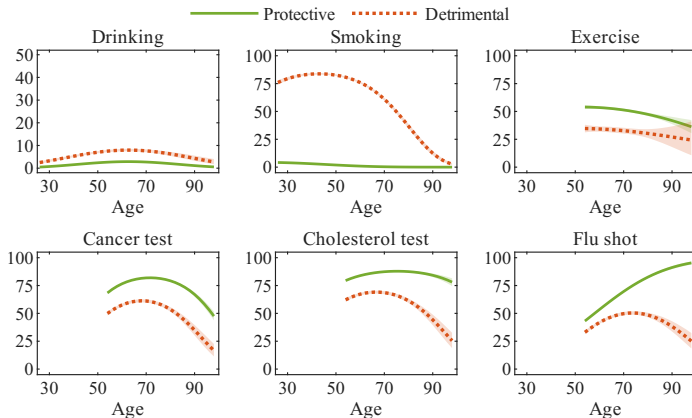
- We assume that observed **health behavior** is the (noisy) result of a **latent time-invariant** factor  $y \in Y \equiv \{y_1, y_2, \dots\}$ 
  - We interpret  $y$  as the individual lifestyle: propensity to engage in healthy behaviors
- We propose a novel econometric model to
  - Allocate individuals to lifestyles  $y$
  - Measure the importance of lifestyles on health dynamics
- We *jointly estimate* health dynamics and lifestyles using a mixture model:

$$\begin{aligned}
 p(\mathbf{z}, \mathbf{h} | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0) &= \sum_{y \in Y} p(\mathbf{z}, \mathbf{h} | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0, y) p(y | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0) \\
 &= \sum_{y \in Y} p(\mathbf{z} | \mathbf{h}, \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0, y) p(\mathbf{h} | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0, y) p(y | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0) \\
 &\simeq \sum_{y \in Y} \underbrace{p(\mathbf{z} | \mathbf{h}, a_0, y)}_{\text{health behavior}} \underbrace{p(\mathbf{h} | \mathbf{s}, \mathbf{e}, a_0, y)}_{\text{health dynamics}} \underbrace{p(y | \mathbf{c}, \mathbf{s}, \mathbf{e}, a_0)}_{\text{lifestyle distribution}}
 \end{aligned}$$

## RESULTS: LIFESTYLES AND HEALTH BEHAVIOR

1. Lifestyles “well” approximated by 2 types: protective and detrimental

### Probability of reporting health behaviors by lifestyle



## RESULTS: LIFESTYLES AND HEALTH DYNAMICS

2. LE at age 50 is 9.0 years larger for **protective** than for **detrimental**

LE at age 50 by education and lifestyles: males born in 1950s

|            | All   |             | HSD   |      | HSG   |      | CG    |      | $\Delta_e \text{LE (CG-HSD)}$ |     |     |
|------------|-------|-------------|-------|------|-------|------|-------|------|-------------------------------|-----|-----|
|            | %     | LE          | %     | LE   | %     | LE   | %     | LE   | Data                          | (a) | (b) |
| All        | 100.0 | <b>27.7</b> | 100.0 | 24.1 | 100.0 | 26.6 | 100.0 | 31.2 | 7.1                           | 4.5 | 2.6 |
| PRO        | 72.1  | <b>30.2</b> | 55.6  | 27.8 | 67.7  | 29.2 | 86.7  | 32.4 | 4.6                           |     |     |
| DET        | 27.9  | <b>21.2</b> | 44.4  | 19.4 | 32.3  | 21.3 | 13.3  | 23.5 | 4.1                           |     |     |
| $\Delta_y$ | 44.2  | <b>9.0</b>  | 11.2  | 8.4  | 35.4  | 7.9  | 73.4  | 8.9  | 0.5                           |     |     |

(a) Gradient explained by difference in health dynamics across education groups for given lifestyle,  $\Delta_e p(\mathbf{h}|\mathbf{e}, y)$

(b) Gradient explained by difference in lifestyles across education groups for given health dynamics,  $\Delta_e p(y|\mathbf{e})$

## RESULTS: LIFESTYLES, EDUCATION, AND HEALTH DYNAMICS

## 3. Lifestyles explain almost 40% of the education gradient of LE

LE at age 50 by education and lifestyles: males born in 1950s

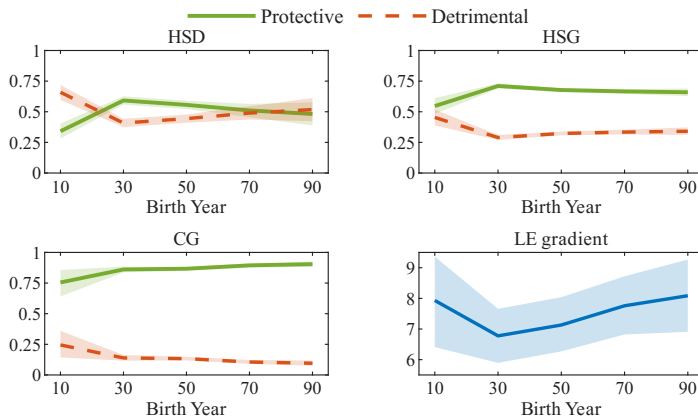
|            | All   |      | HSD         |      | HSG         |      | CG          |      | $\Delta_e \text{LE (CG-HSD)}$ |            |            |
|------------|-------|------|-------------|------|-------------|------|-------------|------|-------------------------------|------------|------------|
|            | %     | LE   | %           | LE   | %           | LE   | %           | LE   | Data                          | (a)        | (b)        |
| All        | 100.0 | 27.7 | 100.0       | 24.1 | 100.0       | 26.6 | 100.0       | 31.2 | <b>7.1</b>                    | <b>4.5</b> | <b>2.6</b> |
| PRO        | 72.1  | 30.2 | <b>55.6</b> | 27.8 | <b>67.7</b> | 29.2 | <b>86.7</b> | 32.4 | 4.6                           |            |            |
| DET        | 27.9  | 21.2 | <b>44.4</b> | 19.4 | <b>32.3</b> | 21.3 | <b>13.3</b> | 23.5 | 4.1                           |            |            |
| $\Delta_y$ | 44.2  | 9.0  | <b>11.2</b> | 8.4  | <b>35.4</b> | 7.9  | <b>73.4</b> | 8.9  | 0.5                           |            |            |

(a) Gradient explained by difference in health dynamics across education groups for given lifestyle,  $\Delta_e p(\mathbf{h}|\mathbf{e}, y)$ (b) Gradient explained by difference in lifestyles across education groups for given health dynamics,  $\Delta_e p(y|\mathbf{e})$

## RESULTS: CHANGES ACROSS COHORTS

## 4. Education gradient of lifestyles widens over time

## Probability of lifestyle at age 50 across cohorts. Males



## The Model

# THE MODEL

## Two different stages

### 1 Early life

- Choice of education and lifestyle

### 2 Life cycle

- Standard life-cycle incomplete-markets model of consumption and wealth accumulation
- Health and labor market risks that are related to **education** and **lifestyle**

Stage 1: Early life

## SET UP

- Teenager/parents in cohort  $c$  make once-and-for-all simultaneous choices of
  - education  $e \in \{HSD, HSG, CG\}$
  - lifestyle  $y \in \{DET, PRO\}$
- They solve  $\max_{e,y} \left\{ V_0^{eyc} - \tau_e - \tau_y \right\}$ 
  - Value  $V_0^{eyc}$  of starting stage 2 with type  $(e, y, c)$
  - Cost  $\tau_e$  of education  $e$ :  $\tau_{HSD} = 0 \quad | \quad \tau_{HSG} = \mu_{HSG} + \sigma_{HSG}\epsilon_e \quad | \quad \tau_{CG} = \mu_{CG} + \sigma_{CG}\epsilon_e$
  - Cost  $\tau_y$  of lifestyle  $y$ :  $\tau_{DET} = 0 \quad | \quad \tau_{PRO} = \mu_{PRO} + \sigma_{PRO}\epsilon_{PRO}$
  - *Correlated* bivariate normal:  $(\epsilon_e, \epsilon_{PRO})' \sim N\left(\mathbf{0}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$
- There are complementarities between education and lifestyle choices if:

$$V_0^{CG,PRO} - V_0^{CG,DET} > V_0^{HSD,PRO} - V_0^{HSD,DET} \Leftrightarrow V_0^{CG,PRO} - V_0^{HSD,PRO} > V_0^{CG,DET} - V_0^{HSD,DET} \quad \text{[theory]}$$

## Stage 2: Life cycle

# MODEL: STATES AND INGREDIENTS

## State variables

### Agent types

- Education:  $e \in \{\text{HSD}, \text{HSG}, \text{CG}\}$
- Lifestyle:  $y \in \{\text{DET}, \text{PRO}\}$
- Cohort:  $c \in \{1930, 1970\}$

### Exogenous states

- Age:  $t \in \{25, 27, 29, \dots\}$
- Health:  $h_t \in \{h_g, h_b\}$
- Earnings shock:  $\zeta_t \in \mathbb{R}$

### Endogenous state

- Cash-on-hand:  $x_t \in [\underline{x}, \infty)$

## Model ingredients

### Cohort-specific

- Employment probability:  $l_t^{\text{ec}}(\varepsilon, h)$
- Labor earnings:  $w_t^{\text{ec}}(\zeta, \varepsilon, h)$

### Cohort-independent

- Health and survival:  $s_t^{\text{ey}}(h), \Gamma_t^{\text{ey}}(h'|h)$
- Medical expenses and insurance:  $m_t^{\text{e}}(\xi, h)$
- Taxes and income floor:  $T(\cdot)$  and  $\underline{x}$
- Utility from consumption and survival

$\Rightarrow$  The solution provides the initial value of each type  $(e, y, c)$ :  $V_0^{\text{eyc}}$

Education gradient of lifestyle

# EDUCATION GRADIENT OF LIFESTYLE

## Mechanisms

- Why higher educated individuals are more likely to be **protective**?

- Direct Effect

① Income:  $w_t^{\text{ec}}(\zeta, \epsilon, h)$  and  $l_t^{\text{ec}}(\epsilon, h)$

⇒ Generates complementarities:  $V_0^{\text{CG,PRO}} - V_0^{\text{CG,DET}} > V_0^{\text{HSD,PRO}} - V_0^{\text{HSD,DET}}$

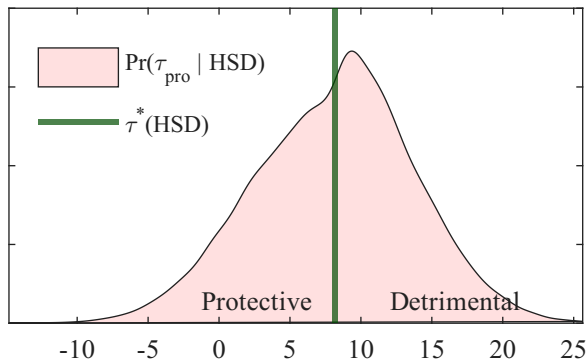
- Induced by complementarities

② Selection:  $F(\tau_{\text{PRO}}|\text{HSD}) \leq F(\tau_{\text{PRO}}|\text{CG}) \quad \forall \tau_{\text{PRO}}$

# EDUCATION GRADIENT OF LIFESTYLE

## Lifestyle Choice for HSD

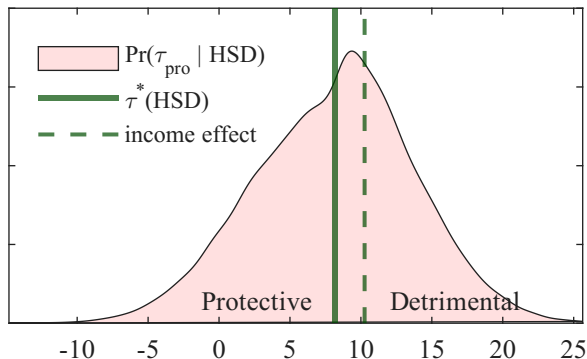
- HSD choose  $y = \text{PRO}$  iff  $\tau_{\text{PRO}} < V_0^{\text{HSD,PRO}} - V_0^{\text{HSD,DET}} \equiv \tau^*(\text{HSD})$



# EDUCATION GRADIENT OF LIFESTYLE

## 1. Income Effect

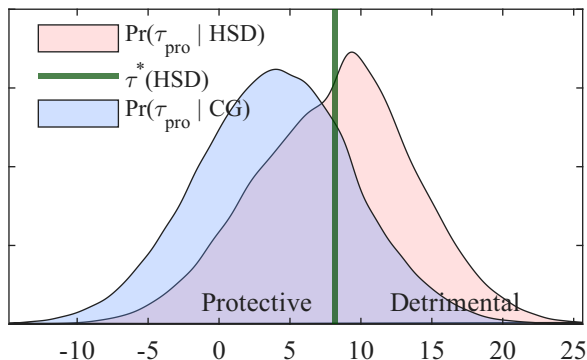
→ If HSD had **same income** prospects as CG: 16pp more of PRO (1.3 more years of LE)



# EDUCATION GRADIENT OF LIFESTYLE

## 2. Selection

→ If HSD had same **distribution of  $\tau_{\text{PRO}}$**  as CG: 31pp more of PRO (2.6 more years of LE)



# EDUCATION GRADIENT OF LIFESTYLE

## Summary

Decomposition: 1970s cohort

|                     | Pr(e) |      | Pr(PRO e) |      |        | LE(e) |      |        |
|---------------------|-------|------|-----------|------|--------|-------|------|--------|
|                     | CG    | HSD  | CG        | HSD  | CG-HSD | CG    | HSD  | CG-HSD |
| Data                | 0.36  | 0.08 | 0.89      | 0.51 | 0.38   | 31.5  | 23.7 | 7.8    |
| Model               | 0.36  | 0.10 | 0.87      | 0.47 | 0.39   | 31.2  | 23.4 | 7.8    |
| Same lifestyle      |       |      | 0.87      | 0.87 | 0.00   | 31.2  | 26.7 | 4.5    |
| Income              |       |      | 0.87      | 0.63 | 0.23   | 31.2  | 24.7 | 6.5    |
| Selection           |       |      | 0.87      | 0.78 | 0.09   | 31.2  | 25.9 | 5.2    |
| No correlated costs | 0.35  | 0.10 | 0.81      | 0.61 | 0.20   | 30.7  | 24.5 | 6.1    |

Policy implications

## POLICY IMPLICATIONS

### Feedback between education and lifestyle choices

- A policy directed at either **early-life investment** can affect the other.
- Given that education and lifestyle choices are **complementary**:
  - Lowering the cost of one investment increases the incentive to undertake the other:
    - 1 Lower cost of **protective behavior**  $\Rightarrow$  healthier lifestyles  $\Rightarrow$  higher educational attainment.
    - 2 Lower **education costs**  $\Rightarrow$  more education  $\Rightarrow$  greater adoption of protective behavior.
- This feedback amplifies the reach of each intervention beyond its direct target.

## POLICY IMPLICATIONS

## Policy counterfactuals

|                    | Pr(e) |       | Pr(PRO) | Pr(PRO e) |       |        | LE(e) |      |      |        |
|--------------------|-------|-------|---------|-----------|-------|--------|-------|------|------|--------|
|                    | CG    | HSD   |         | CG        | HSD   | CG-HSD | All   | CG   | HSD  | CG-HSD |
| Benchmark          | 0.36  | 0.10  | 0.73    | 0.87      | 0.47  | 0.39   | 28.0  | 31.2 | 23.4 | 7.8    |
| Cost pro - 0.5 sd  | 0.36  | 0.09  | 0.86    | 0.95      | 0.67  | 0.27   | 29.1  | 31.9 | 25.1 | 6.8    |
| Diff pro           | 0.01  | -0.01 | 0.14    | 0.08      | 0.20  | -0.12  | 1.1   | 0.7  | 1.7  | -1.0   |
| Cost educ - 0.5 sd | 0.55  | 0.04  | 0.74    | 0.84      | 0.42  | 0.42   | 28.8  | 31.0 | 22.9 | 8.1    |
| Diff educ          | 0.19  | -0.06 | 0.02    | -0.02     | -0.05 | 0.03   | 0.8   | -0.2 | -0.4 | 0.2    |

## Conclusions

# CONCLUSIONS

- We develop an econometric model to identify latent types in lifestyles
  - ⇒ Differences in lifestyles across education groups account for 40% of the LE gradient (2.6 out of 7.1 years)
- HA model w/ complementarities in education and lifestyle investments
  - ① Education gradient of lifestyles
    - Income advantage explain 40% (⇒ 1.3 years)
    - Endogenous + Primitive selection explains 75% (⇒ 2.6 year)
    - Both similarly important
    - Not explained by lifestyles (⇒ 4.5 years)
  - ② 1.2 years increase in education gradient of LE between 1930s and 1970s cohorts
    - Direct effect of wage changes explains 4.8 months
    - Indirect effect through selection explains 9.6 months
    - Both together explain 1.2 year

# ECONOMETRIC MODEL

## 1. Health Behavior

- How to model  $p(\mathbf{z}|\mathbf{h}, a_0, y)$ ?
- We assume that, conditional on  $(y_i, h_{it}, a_{it})$  health behaviors  $z_{m,it}$  are
  - independent from  $h_{it'} \forall t' \neq t$
  - independent between them
  - independent over time
- Hence, we can write the probability of a given history  $\mathbf{z}_i$  of health behaviors as

$$p(\mathbf{z}_i|\mathbf{h}_i, y_i, a_{it}) = \prod_t \prod_m p(z_{m,it} | y_i, h_{it}, a_{it})$$

- We model each element  $p(z_{m,it}|y_i, h_{it}, a_{it})$  as a probit

Back to Mixture Model

# ECONOMETRIC MODEL

## 2. Health Dynamics

- How to model  $p(\mathbf{h}|s, e, a_0, y)$ ?
- We assume that, conditional on  $(s, e, a_0, y)$ , the evolution of health outcomes is markovian
- Hence, we can write the probability of a given history  $\mathbf{h}_i$  of health outcomes as

$$p(\mathbf{h}_i | s_i, e_i, a_{it}, y_i) = \prod_t p(h_{it} | s_i, e_i, a_{it}, y_i, h_{i,t-1})$$

- We model each element  $p(h_{it} | s_i, e_i, a_{it}, y_i, h_{i,t-1})$  as a nested probit
  - First we model survival
  - Next, we model health changes conditional on survival

Back to Mixture Model

# ECONOMETRIC MODEL

## 3. Weights

- How to model the mixture weights  $p(y|c, s, e, a_0)$ ?
- The conditional mixture weights are different at each age of entry  $a_0$  because of differential health dynamics by type  $y$ 
  - **Detrimental** lifestyles become less frequent at older ages because they die.
- What we do
  - Use the health dynamics to express  $p(y|c_i, s_i, e_i, a_{it}, h_{it})$  as a function of  $p(y|c_i, s_i, e_i, a_{it-1}, h_{it-1})$
  - Estimate the mixture weights at the initial age of 25,  $p(y|c_i, s_i, e_i, 25, h_{i,25})$
- We model each element  $p(y|c_i, s_i, e_i, 25, h_{i,25})$  as a multinomial probit

Back to Mixture Model

## HEALTH BEHAVIOUR DATA

**Mean health behavior and 4-year auto-correlation**

|             | Mean |      |      |       |       | AC    |       |
|-------------|------|------|------|-------|-------|-------|-------|
|             | HSD  | HSG  | CG   | 65-70 | 75-80 | 65-70 | 75-80 |
| Drinking    | 0.07 | 0.08 | 0.07 | 0.08  | 0.04  | 0.46  | 0.39  |
| Smoking     | 0.18 | 0.15 | 0.07 | 0.14  | 0.08  | 0.70  | 0.64  |
| Cancer test | 0.60 | 0.71 | 0.79 | 0.77  | 0.72  | 0.40  | 0.41  |
| Cholesterol | 0.78 | 0.85 | 0.89 | 0.84  | 0.85  | 0.30  | 0.27  |
| Flu shot    | 0.67 | 0.74 | 0.81 | 0.66  | 0.76  | 0.59  | 0.59  |
| Exercise    | 0.27 | 0.38 | 0.54 | 0.41  | 0.38  | 0.41  | 0.37  |

Data from the HRS. HSD: high school dropout; HSG: high school graduate; CG: college graduate; 65-70: sub-sample of individuals aged 65 to 70; 75-80: sub-sample of individuals aged 75 to 80. The last two columns show the autocorrelation (AC) of each health behavior with a 4-year lag.

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## HEALTH BEHAVIOUR DATA

Cross correlation health behaviors

|             | Drinking | Smoking | Cancer test | Cholesterol | Flu shot | Exercise |
|-------------|----------|---------|-------------|-------------|----------|----------|
| Drinking    | 1.00     | 0.07    | 0.00        | -0.00       | -0.03    | 0.02     |
| Smoking     | 0.12     | 1.00    | -0.08       | -0.07       | -0.08    | -0.06    |
| Cancer test | -0.02    | -0.15   | 1.00        | 0.28        | 0.19     | 0.10     |
| Cholesterol | -0.02    | -0.12   | 0.36        | 1.00        | 0.22     | 0.07     |
| Flu shot    | -0.05    | -0.09   | 0.21        | 0.24        | 1.00     | 0.03     |
| Exercise    | 0.03     | -0.09   | 0.11        | 0.03        | 0.01     | 1.00     |

Data from the HRS. Lower diagonal: individuals aged between 65 and 70. Upper diagonal: individuals aged between 75 and 80.

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## PROPOSITION

- Let the distributions of  $\epsilon_e$  and  $\epsilon_{\text{PRO}}$  be independent
- Let  $F(\tau_{\text{PRO}}|e)$  be the CDF of  $\tau_{\text{PRO}}$  conditional on education choice  $e$
- Let's consider only two education choices  $e \in \{\text{HSD}, \text{CG}\}$
- Proposition:
  - If the choices of education and lifestyle are complementary, the distribution  $F(\tau_{\text{PRO}}|\text{HSD})$  first-order stochastically dominates  $F(\tau_{\text{PRO}}|\text{CG})$ . That is,

$$F(\tau_{\text{PRO}}|\text{HSD}) \leq F(\tau_{\text{PRO}}|\text{CG}) \quad \forall \tau_{\text{PRO}}$$

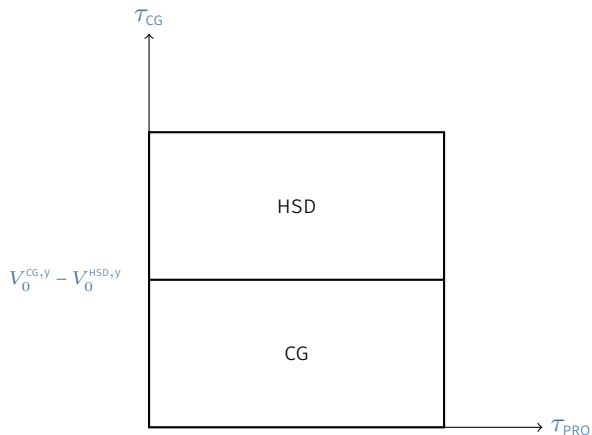
with strict inequality for some  $\tau_{\text{PRO}}$ .

- Instead, if the choices of education and lifestyle are independent we have that,

$$F(\tau_{\text{PRO}}|\text{HSD}) = F(\tau_{\text{PRO}}|\text{CG}) \quad \forall \tau_{\text{PRO}}$$

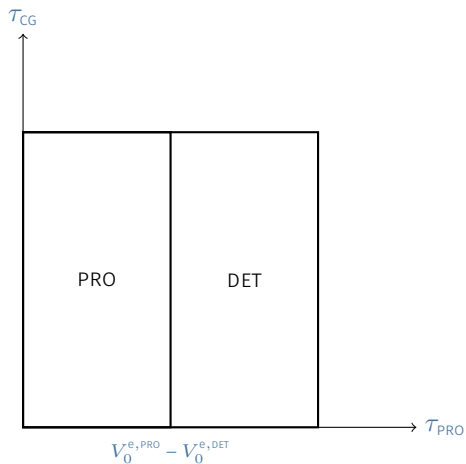
## CHOICES: TWO EDUCATION LEVELS

Independent choices

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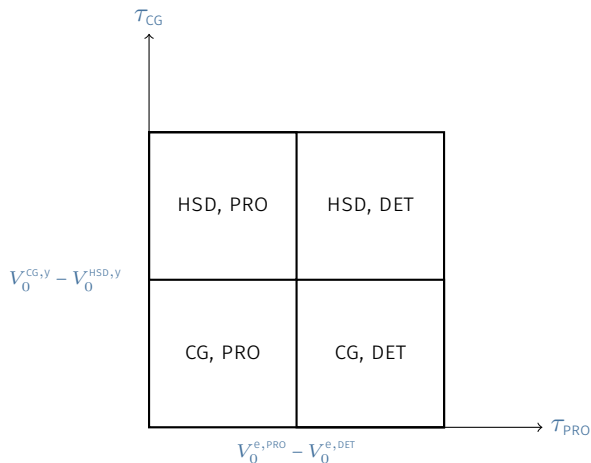
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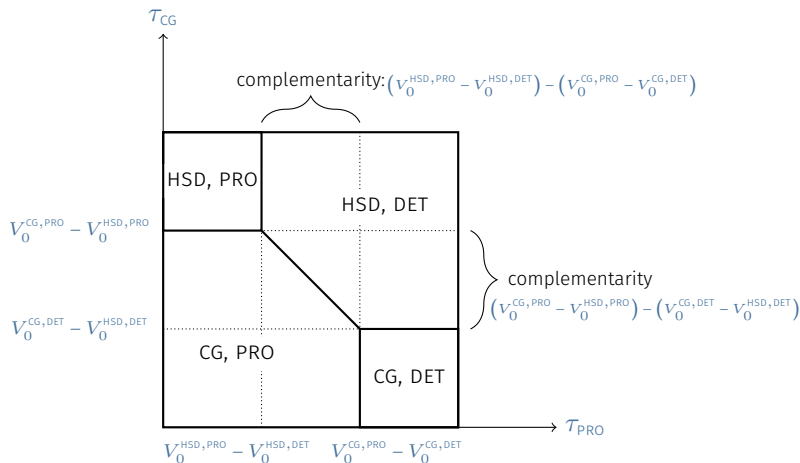
## CHOICES: TWO EDUCATION LEVELS

Independent choices


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## CHOICES: TWO EDUCATION LEVELS

## Complementarities


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